

Serendipity: See Koestler (1959) and Rees (2004). Rees also has powerful ideas on fore-castability. See also Popper’s comments in Popper (2002), and Waller (2002a), Cannon (1940), Mach (1896) (cited in Simonton [1999]), and Merton and Barber (2004). See Simonton (2004) for a synthesis. For serendipity in medicine and anesthesiology, see Vale et al. (2005).

“Renaissance man”: See www.bell-labs.com/project/feature/archives/cosmology/.

Laser: As usual, there are controversies as to who “invented” the technology. After a successful discovery, precursors are rapidly found, owing to the retrospective distortion. Charles Townsend won the Nobel prize, but was sued by his student Gordon Gould, who held that he did the actual work (see *The Economist*, June 9, 2005).

Darwin/Wallace: Quammen (2006).

Popper’s attack on historicism: See Popper (2002). Note that I am reinterpreting Popper’s idea in a modern manner here, using my own experiences and knowledge, not commenting on comments about Popper’s work—with the consequent lack of fidelity to his message. In other words, these are not directly Popper’s arguments, but largely mine phrased in a Popperian framework. The conditional expectation of an unconditional expectation is an unconditional expectation.

Forecast for the future a hundred years earlier: Bellamy (1891) illustrates our mental projections of the future. However, some stories might be exaggerated: “A Patently False Patent Myth still! Did a patent official really once resign because he thought nothing was left to invent? Once such myths start they take on a life of their own.” *Skeptical Inquirer*, May–June, 2003.

Observation by Peirce: Olsson (2006), Peirce (1955).

Predicting and explaining: See Thom (1993).

Poincaré: The three body problem can be found in Barrow-Green (1996), Rollet (2005), and Galison (2003). On Einstein, Pais (1982). More recent revelations in Hladik (2004).

Billiard balls: Berry (1978) and Pisarenko and Sornette (2004).

Very general discussion on “complexity”: Benkirane (2002), Scheps